Downward-sloping graph

Graphs of linear equations are read from left to right and slope down or up. Downward-sloping graphs have a negative gradient; upward-sloping ones have a positive gradient.

The equation here contains the term –2x. Because x is multiplied by a negative number (–2), the graph will slope downward.

this means x multiplied by –2

$$y = -2x + 1$$

Use a table to find some values for x and y. This equation is more complex than the last, so add more rows to the table: –2x and 1. Calculate each of these values, then add them to find y. It is important to keep track of negative signs in front of numbers.

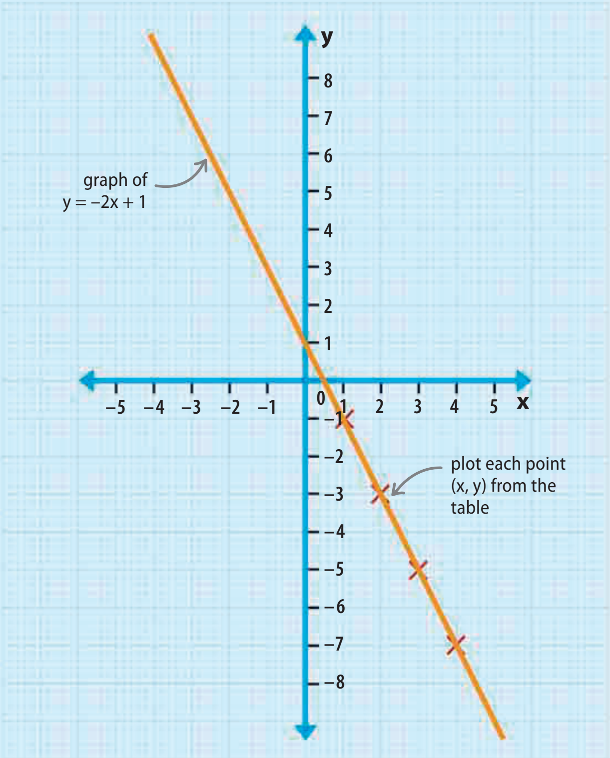
x	–2x	+1	y=–2x+1
1	–2	+1	–1
2	–4	+1	–3
3	–6	+1	–5
4	–8	+1	–7

write down some possible values of x

work out corresponding values for y by adding together the parts of the equation

values of x multiplied by –2

+1 is constant



REAL WORLD

Temperature conversion graph

A linear graph can be used to show the conversion between the two main methods of measuring temperature—Fahrenheit and Celsius. To convert any temperature from Fahrenheit into Celsius, start at the position of the Fahrenheit temperature on the y axis, read horizontally across to the line, and then vertically down to the x axis to find the Celsius value.

°F	°C
32.0	0
50.0	10

△ **Temperature conversion**
Two sets of values for Fahrenheit (F) and Celsius (C) give all the information that is needed to plot the conversion graph.